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SIMATS
ENGINEERING



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ITA0502-COMPUTER VISION

MCQ TEST

1. At a particular pixel, $I_x = 4$ and $I_y = 3$. What are the values of I_x^2 and I_y^2 ?
A) 8, 6 B) ~~16, 9~~ C) 12, 7 D) 4, 3
2. At a particular pixel, $I_x = 5$ and $I_y = 2$. What is the value of $I_x \times I_y$?
A) 7 B) ~~10~~ C) 25 D) 4
3. For a window with I_x values [1, 2, 3], what is the value of $\text{Sum}(I_x^2)$?
A) 6 B) 9 C) ~~14~~ D) 36
4. Given the Harris matrix $M = \begin{bmatrix} 10 & 4 \\ 4 & 8 \end{bmatrix}$, what is the determinant of M ?
A) ~~64~~ B) 72 C) 80 D) 96
5. Given the Harris matrix $M = \begin{bmatrix} 12 & 3 \\ 3 & 8 \end{bmatrix}$, what is the trace of M ?
A) 5 B) 12 C) ~~20~~ D) 24
6. What is the main benefit of image enhancement?
A) ~~To make important image features easier to identify~~
B) To remove all image information
C) To increase image distortion
D) To reduce the number of objects in an image
7. An 8-bit grayscale image can represent how many different intensity levels?
A) 8 B) 128 C) ~~256~~ D) 512
8. Which of the following best represents a grayscale image?
A) Each pixel contains only red, green, and blue components
B) ~~Each pixel represents an intensity value~~
C) Each pixel contains only text information
D) Each pixel represents a geometric shape

9. If $\text{Sum}(I_x^2) = 20$, $\text{Sum}(I_y^2) = 10$, and $\text{Sum}(I_x I_y) = 4$, what is the Harris response R?

$$\begin{bmatrix} 25 & 4 \\ 4 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 4 \\ 4 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 16 & 4 \\ 4 & 25 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 25 \\ 16 & 4 \end{bmatrix}$$

10. If $\text{determinant}(M) = 100$, $\text{trace}(M) = 20$, and $k = 0.04$, what is the Harris response R?

11. If $\text{Sum}(I_x^2) = 25$, $\text{Sum}(I_y^2) = 16$, $\text{Sum}(I_x I_y) = 4$, and $k = 0.04$, what is the approximate Harris response R?

12. For an image point $(x, y) = (4, 3)$, calculate rho using $f = x \cos(\theta) + y \sin(\theta)$ when $\theta = 0$ degrees.

13. For an image point $(x, y) = (2, 5)$, calculate rho when $\theta = 90$ degrees.

14. Three edge points belonging to the same straight line vote for the same parameter pair (ρ, θ) . What will be the accumulator value at that location?

15. A Hough accumulator contains the following voting values: A = 3, B = 12, C = 5, D = 2. If the minimum threshold for detecting a line is 5 votes, which parameter combinations represent possible lines?

16. A straight line in an image contains some missing edge pixels. The Hough Transform is still able to detect the line because:

- A) It requires every pixel of the line
- B) The available edge pixels vote for common line parameters
- C) It automatically fills all missing pixels
- D) It removes the missing pixels before voting

17. For a point $(x, y) = (5, 4)$, the Hough Transform is evaluated at $\theta = 90$ degrees. What is the value of ρ ?

- ☒ A) 4 ☐ B) 5 ☐ C) 9 ☐ D) 20

18. A Hough accumulator has the following values: 2, 4, 15, 3, and 6. Which value indicates the strongest candidate for a straight line?

- ☐ A) 2 ☐ B) 4 ☐ C) 6 ☒ D) 15

19. Four edge points produce the following votes for three parameter combinations: Parameter A = 2 votes, Parameter B = 8 votes, Parameter C = 5 votes. If the detection threshold is 5 votes, how many lines are detected?

- ☐ A) 1 ☒ B) 2 ☐ C) 3 ☐ D) 0

20. Which of the following is the main purpose of image preprocessing?

- ☒ A) To improve image quality before further processing
☐ B) To increase image noise
☐ C) To reduce the image information
☐ D) To convert an image into a video

21. Which of the following is a common image preprocessing technique used to reduce noise?

- ☒ A) Smoothing ☐ B) Hough Transform ☐ C) Corner Detection ☐ D) Edge Linking

22. Which type of image processing operates directly on the pixel values of an image?

- ☒ A) Spatial domain processing
☐ B) Frequency domain processing
☐ C) Feature matching
☐ D) Object recognition

23. What is an edge in a digital image?

- ☒ A) A region with constant intensity
☐ B) A location where there is a significant change in intensity
☐ C) A completely uniform region
☐ D) A region containing only noise

24. Which technique is specifically designed to identify points where intensity changes significantly in multiple directions?

- ☒ A) Smoothing ☐ B) Harris Corner Detection ☐ C) Histogram Processing
☐ D) Image Compression

25. Why are corners considered important features in image processing?

- ☒ A) They provide distinctive points that can be used for feature detection and matching
☐ B) They always have the same intensity
☐ C) They occur only in binary images
☐ D) They are used only for noise removal